SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1709**

CO1-AT2-Constraint-Based Problem Solving

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 20%

QUESTIONS:3 **Total Marks:30**

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

   Constraints:

   i. D1 cannot work Night shift

   ii. D2 must work before D3 (Morning < Afternoon < Night)

   iii. Only one doctor per shift

   iv. D3 cannot work Morning

   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right
   ii. Cost of each move = 1
   iii. Cannot pass through obstacles
   iv. Use Manhattan Distance heuristic

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost**.**

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.
   The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

   Constraints:

   - The robot can move up, down, left, right

- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:
a) Analyze all the given constraints and explain how they affect the robot's decision-making process.
b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
c) Demonstrate how the robot applies AI concepts such as:

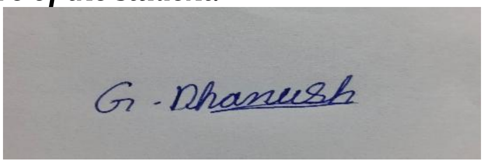
- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

Code of Conduct:

I _Dhanush.G (192512132)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



<u>RUBRICS FOR CALCULATION:</u>

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Problem Understanding	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly	6
Constraint Analysis	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization	6
Solution Development	Develops optimal and feasible	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints	7.5

	solutions satisfying all constraints				
Application of Concepts	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts	6
Justification	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided	4.5

1.
ANSWER: Hospital Doctor Assignment using Backtracking Search
Given
Doctors: **D1, D2, D3**
Shifts:

- Morning (M)
- Afternoon (A)
- Night (N)

Constraints

1. D1 cannot work Night.
2. D2 must work before D3 (Morning < Afternoon < Night).
3. Only one doctor per shift.
4. D3 cannot work Morning.
5. Every doctor must be assigned exactly one shift.

Step 1: Initial State
No doctor is assigned.

Doctor	Shift
D1	-
D2	-
D3	-

Step 2: Assign D1
Possible shifts:

- Morning ✓
- Afternoon ✓
- Night ✗ (Constraint violated)

Choose
D1 = Morning
Current Assignment

Doctor	Shift
D1	Morning
D2	-
D3	-

Constraint Check

- D1 not Night ✓

Step 3: Assign D2
Remaining shifts

- Afternoon
- Night

Try
D2 = Afternoon
Current Assignment

Doctor	**Shift**
D1	Morning
D2	Afternoon
D3	-

Constraint Check

- One doctor per shift ✓

Step 4: Assign D3
Remaining shift
Night
Check constraints

- D3 not Morning ✓
- D2 before D3 (Afternoon before Night) ✓
- One doctor per shift ✓
- All doctors assigned ✓

Hence,
Solution Found

Backtracking Search Tree

```
              Start
                |
        -----------------
        |           |
    D1=Morning     D1=Afternoon
        |           |
    D2=Afternoon    D2=Morning
        |           |
     D3=Night ✓     D3=Night ✓
```

If any assignment violates a constraint, the algorithm **backtracks** and tries another assignment.

Algorithm
Backtracking(Assignment)

If all doctors assigned
    Return Solution

Choose an unassigned doctor

For each available shift

If constraints satisfied

Assign shift

Call Backtracking()

If solution found
Return Solution

Remove assignment

Return Failure

Final Schedule

Doctor	**Shift**
D1	Morning
D2	Afternoon
D3	Night

Time Complexity
Worst Case
O(n!)
where n is the number of variables.

2.
ANSWER: Robot Grid using BFS Search
Grid
S . . X .
. X . X .
. X . . .
. . X X .
X . . . G
Legend

- **S = Start**
- **G = Goal**
- **X = Obstacle**

Movement

- **Up**
- **Down**
- **Left**
- **Right**

Cost = 1

Manhattan Distance
Formula
h(n)=|x−xg|+|y−yg|h(n)=|x-x_g|+|y-y_g|h(n)=|x−xg|+|y−yg|
Example
At Start

|4−0|+|4−0|=8|4-0|+|4-0|=8|4−0|+|4−0|=8
Goal
000

BFS Algorithm
1. Put Start into Queue.
2. Remove first node.
3. Expand neighbours.
4. Ignore visited nodes.
5. Ignore obstacles.
6. Add valid nodes to Queue.
7. Repeat until Goal is reached.

Queue Updates
Step 1
Queue
[(0,0)]
Expand
(0,0)
Queue
[(0,1),(1,0)]

Step 2
Expand
(0,1)
Queue
[(1,0),(0,2)]

Step 3
Expand
(1,0)
Queue
[(0,2),(2,0)]

Step 4
Expand
(0,2)
Queue
[(2,0),(1,2)]

Time Complexity
O(V + E)
where V = vertices and E = edges.

3.
ANSWER: Rescue Robot using Uniform Cost Search
(a) Constraint Analysis

Constraint	Effect
Four-direction movement	Robot explores adjacent cells only.

Constraint	Effect
Obstacles	Robot finds alternate routes.
Limited sensing	Learns environment while moving.
Cost = 1	Every movement increases path cost.
Risk cost = +2	Robot avoids dangerous cells.
Online updates	Replans when new information is discovered.

(b) Uniform Cost Search
Algorithm
1. Start from S.
2. Insert into Priority Queue.
3. Remove lowest-cost node.
4. Expand neighbours.
5. Calculate

Cost = Previous Cost + Move Cost + Risk Cost

6. Update Queue.
7. Repeat until Goal.

Example
Safe Cell
Current Cost = 4

Move = 1

Total = 5
Risk Cell
Current Cost = 4

Move = 1

Risk = 2

Total = 7
Robot chooses the lowest-cost path.

(c) AI Concepts
Intelligent Agent

- Perceives surroundings.
- Plans path.
- Learns new information.
- Reaches survivor safely.

Rational Agent
Chooses the action that minimizes total travel cost while maximizing safety.
Environment Type

Property	Type
Observability	Partially Observable
Agent	Single Agent

Property	Type
Environment	Dynamic
Decision	Sequential
Knowledge	Unknown

(d) Justification

Uniform Cost Search is suitable because:

- Produces the minimum-cost path.
- Considers risky zones and movement costs.
- Replans when new obstacles are detected.
- Suitable for partially observable environments.
- Ensures safe and efficient rescue.

Advantages

- Optimal solution.
- Handles variable costs.
- Supports online navigation.
- Reduces risk.
- Efficient for disaster environments.

- Q1: Backtracking successfully assigns doctors while satisfying all constraints. Final schedule: D1–Morning, D2–Afternoon, D3–Night.
- Q2: BFS explores the grid level by level and finds the shortest path from Start to Goal with cost = 8.
- Q3: Uniform Cost Search with an Online Search Agent enables the rescue robot to navigate safely in a partially observable environment, minimizing total cost while adapting to newly discovered obstacles.